King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam II Term 241

Original exam date: 5 November 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Consider the subspace S of $\mathbb{R}^3$ defined by

$$S = \{(x, y, z) \mid x - 4y + 7z = 0\}.$$

A basis of S consists of the vectors

(a) $\mathbf{v}_1 = (4, 1, 0)$, $\mathbf{v}_2 = (-7, 0, 1)$

(b) $\mathbf{v}_1 = (1, 1, 0)$, $\mathbf{v}_2 = (-7, 0, 1)$

(c) $\mathbf{v}_1 = (4, 2, 0)$, $\mathbf{v}_2 = (7, 0, 1)$

(d) $\mathbf{v}_1 = (-4, 2, 0)$, $\mathbf{v}_2 = (-7, 0, 1)$

(e) $\mathbf{v}_1 = (4, -1, 0)$, $\mathbf{v}_2 = (-7, 0, 1)$

2. The eigenvector associated with the eigenvalue $\lambda = 0$ of the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -2 & -1 \\ -2 & 6 & 3 \end{bmatrix}$$

is $\begin{bmatrix} \alpha \\ -1 \\ \beta \end{bmatrix}$, then $\alpha + \beta =$

(a) -3

(b) 2

(c) -4

(d) 4

(e) 0

3. A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = Ae^{2x} + B\cos(2x) + C\sin(2x)$$

is

(a) $y''' - 2y'' + 4y' - 8y = 0$

(b) $y''' - 2y'' + 4y' + 8y = 0$

(c) $y''' - 2y'' - 4y' - 8y = 0$

(d) $y''' - 3y'' + 4y' + 8y = 0$

(e) $y''' + 2y'' + 4y' - 8y = 0$

4. The rank of the matrix

$$A = \begin{bmatrix} 1 & -3 & -8 & -5 \\ 2 & 1 & -4 & 11 \\ 1 & 3 & 3 & 13 \end{bmatrix}$$

is

(a) 3

(b) 0

(c) 2

(d) 1

(e) 4

5. If the matrix

$$A = \begin{bmatrix} 6 & -10 \\ 2 & -3 \end{bmatrix}$$

is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 2 & 5 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 5 & 2 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$

(e) $P = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

6. If $W(x)$ is the Wronskian of the functions $f(x) = e^x \sin x$ and $g(x) = e^x \cos x$, then $W(x) =$

(a) $-e^{-3x}$

(b) e^{4x}

(c) 2

(d) e^{3x}

(e) $-e^{2x}$

7. A particular solution of the differential equation

$$y'' + y = \sec x$$

is given by $y_p(x) =$

(a) $2x \sin x + \cos x \ln |\sin x|$

(b) $x \sin x + \cos x \ln |\cos x|$

(c) $x^2 \sin x + \cos x \ln |\sec x|$

(d) $x^2 \sin x + \sin x \ln |\cos x|$

(e) $\sin x + \cos x \ln |\cos x|$

8. The general solution of the differential equation

$$y^{(4)} + y^{(3)} - 3y'' - 5y' - 2y = 0$$

is

(a) $y(x) = c_1 e^{-2x} + c_2 e^{3x} + (c_3 + c_4 x)e^{-x}$

(b) $y(x) = c_1 e^{2x} + (c_2 + c_3 x + c_4 x^2)e^{-x}$

(c) $y(x) = c_1 e^{2x} + c_2 e^{-2x} + (c_3 + c_4 x)e^{-x}$

(d) $y(x) = c_1 e^{-2x} + (c_2 + c_3 x + c_4 x^2)e^{-x}$

(e) $y(x) = c_1 e^{2x} + (c_2 + c_3 x + c_4 x^2)e^{x}$

9. If the characteristic polynomial of the matrix

$$A = \begin{bmatrix} 7 & -3 & 1 \\ 8 & -3 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

is $p(\lambda) = -(\lambda - 1)(\lambda - 3)^2$, then a basis for the eigenspace of $\lambda = 3$ is

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ \beta \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ \alpha \end{bmatrix},$$

then $\alpha - \beta =$

(a) 6

(b) -4

(c) 0

(d) -8

(e) 2

10. The characteristic polynomial of the matrix

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 4 & -4 & -2 \\ -2 & 12 & 6 \end{bmatrix}$$

is $p(\lambda) =$

(a) $-\lambda^3 + 7\lambda^2 + 10\lambda$

(b) $-\lambda^3 + 7\lambda^2 - 8\lambda + 4$

(c) $-\lambda^3 - 7\lambda^2 - 10\lambda$

(d) $-\lambda^3 + 7\lambda^2 - 10\lambda$

(e) $-\lambda^3 - 7\lambda^2 - 8\lambda + 2$

11. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$y^{(5)} - y' = (1 + 2x)e^{-x} + 3$$

is given by $y_p(x) =$

(a) $Ax + (Bx + Cx^2)e^{-x}$

(b) $Ax + (B + Cx)e^{-x}$

(c) $A + (B + Cx)e^{-x}$

(d) $Ax^2 + (Bx + Cx^2)e^{-x}$

(e) $A + (Bx + Cx^2)e^{-x}$

12. If the solution space of the system

$$\begin{aligned} x_1 + 2x_2 + 7x_3 - 9x_4 + 31x_5 &= 0 \\ 2x_1 + 4x_2 + 7x_3 - 11x_4 + 34x_5 &= 0 \\ 3x_1 + 6x_2 + 5x_3 - 11x_4 + 29x_5 &= 0 \end{aligned}$$

has all linear combinations of the three vectors $\mathbf{u} = (\alpha, 1, 0, 0, 0)$, $\mathbf{v} = (2, 0, 1, \beta, 0)$ and $\mathbf{w} = (-3, 0, \gamma, 0, 1)$, then $\alpha + \beta + \gamma =$

(a) 6

(b) -4

(c) -5

(d) 0

(e) 4

13. If $y_p = Ae^x \cos(x) + Be^x \sin(x)$ is a particular solution of the differential equation

$$y'' + 2y' + 5y = e^x \sin x,$$

then $65A + 65B =$

(a) 3

(b) 11

(c) -4

(d) 4

(e) 5

14. If $y(x)$ is the solution of the initial-value problem

$$y'' + 2y' + y = 0; \qquad y(0) = 2, \quad y'(0) = -1,$$

then $y(1) =$

(a) $\frac{2}{e}$

(b) $\frac{4}{e}$

(c) $\frac{3}{e}$

(d) 0

(e) $\frac{1}{e}$

15. The general solution of the differential equation

$$y'' + 2y' + 5y = 0$$

is

(a) $y(x) = c_1 e^{2x} \sin(x) + c_2 e^{2x} \cos(x)$

(b) $y(x) = c_1 e^{x} \sin(2x) + c_2 e^{x} \cos(2x)$

(c) $y(x) = c_1 e^{-x} \sin(2x) + c_2 e^{-x} \cos(2x)$

(d) $y(x) = c_1 e^{-2x} \sin(x) + c_2 e^{-2x} \cos(x)$

(e) $y(x) = c_1 e^{-2x} \sin(2x) + c_2 e^{2x}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$x = 4y - 7z$: $(x, y, z) = y(4, 1, 0) + z(-7, 0, 1)$.
2	**(b)**	$A(\alpha, -1, \beta) = \mathbf{0}$: $\alpha = 0$, $\beta = 2$, sum 2.
3	**(a)**	Roots $2, \pm 2i$: $(r-2)(r^2+4) = r^3 - 2r^2 + 4r - 8$.
4	**(a)**	Echelon rows $(1, -3, -8, -5)$, $(0, 7, 12, 21)$, $(0, 0, 5, 0)$: rank 3.
5	**(e)**	$\lambda = 1$: $(2, 1)$; $\lambda = 2$: $(5, 2)$; $P = [\,(2,1)\ (5,2)\,]$, $D = \mathrm{diag}(1, 2)$.
6	**(e)**	$W = fg' - f'g = e^{2x}(-\sin^2 x - \cos^2 x) = -e^{2x}$.
7	**(b)**	Variation of parameters, $W = 1$: $u_1 = \ln\lvert\cos x\rvert$, $u_2 = x$.
8	**(b)**	$r^4 + r^3 - 3r^2 - 5r - 2 = (r-2)(r+1)^3$: $y = c_1 e^{2x} + (c_2 + c_3 x + c_4 x^2)e^{-x}$.
9	**(c)**	Eigenspace: $4x - 3y + z = 0$, so $\beta = 4$, $\alpha = 4$, and $\alpha - \beta = 0$.
10	**(d)**	$(5 - \lambda)(\lambda^2 - 2\lambda) = -\lambda^3 + 7\lambda^2 - 10\lambda$.
11	**(a)**	Roots $0, \pm 1, \pm i$ (simple): both parts get a factor x, $y_p = Ax + (Bx + Cx^2)e^{-x}$.
12	**(c)**	$x_3 = x_4 - 4x_5$, $x_1 = -2x_2 + 2x_4 - 3x_5$: $\alpha = -2$, $\beta = 1$, $\gamma = -4$, sum -5.
13	**(a)**	$7A + 4B = 0$, $-4A + 7B = 1$: $A = -\frac{4}{65}$, $B = \frac{7}{65}$, so $65(A + B) = 3$.
14	**(c)**	$r = -1$ (double): $y = (2 + x)e^{-x}$, so $y(1) = \frac{3}{e}$.
15	**(c)**	$r^2 + 2r + 5 = 0 \Rightarrow r = -1 \pm 2i$: $y = e^{-x}(c_1 \sin 2x + c_2 \cos 2x)$.